



A Project Synopsis of

BUDGET PLANNING SYSTEM

*Submitted for the partial fulfillment of the requirement for
the degree of*

Bachelor of Computer Applications (BCA)

Academic Session- 2023-2026

Department of Computer Applications

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude and heartfelt thanks to all those who have directly or indirectly contributed to the successful completion of our project titled "**Budget Planning System**", submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA)** during the academic session **2023–2026**. This project would not have been possible without the continuous support, guidance, and encouragement of many respected individuals.

First and foremost, we express our deep sense of gratitude to our respected **Project Guide, Mr. Ritesh Jha**, for his valuable guidance, constant encouragement, constructive suggestions, and continuous support throughout the development of this project. His knowledge, patience, and motivation helped us to understand the project requirements clearly and complete the work successfully within the stipulated time.

We would also like to extend our sincere thanks to our respected **Chancellor Sir, Registrar Sir, Dean Sir and Head of the Department (HOD)** for their leadership, administrative support, and encouragement. Their vision, academic guidance, and commitment to providing quality education created a positive learning environment that enabled us to complete this project successfully. We are also thankful to all faculty members and staff of the Department of Computer Applications for providing the necessary facilities, cooperation, and valuable suggestions during the preparation of this project.

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DECLARATION

We hereby declare that the project entitled "Budget Planning System" submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) to NSU, Jamshedpur, is a record of original work carried out by us under the guidance and supervision of Dr. Anand Mishra, Project Guide, Department of Computer Applications.

We further declare that this project work has been completed by us independently and has not been submitted, either in part or in full, to any other university or institution for the award of any degree, diploma, or certificate.

We also confirm that all the information, data, and materials used in this project have been properly acknowledged and referenced wherever required. We have complied with all academic rules, regulations, and ethical standards prescribed by the institution. Any assistance received during the development of this project has been duly acknowledged in the acknowledgement section of this report.

We understand that any act of plagiarism or academic dishonesty, if found at any stage, may lead to cancellation of our project and disciplinary action as per the rules of the university.

This project represents our genuine effort and learning during the course of study in the Bachelor of Computer Applications (BCA) program for the academic session 2023–2026.

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TITLE OF THE PROJECT**BUDGET PLANNING SYSTEM**

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Academic Session: 2023 – 2026

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INTRODUCTION

The **Budget Planning System** is a web-based application designed to help individuals, families, and organizations manage their finances effectively by tracking income, expenses, and savings goals. In today's rapidly changing economic environment, proper financial management has become an essential skill for personal as well as professional growth. The increasing complexity of financial transactions, rising costs of living, and the need for long-term savings have created a strong demand for intelligent and user-friendly tools that can assist users in planning and controlling their budgets efficiently.

In traditional budget management, individuals usually maintain handwritten records, use paper-based ledgers, or rely on basic spreadsheets to track their financial activities. These manual methods are time-consuming, error-prone, and often fail to provide meaningful insights into spending patterns and financial health. People frequently overspend in certain categories without realizing it, miss bill payment deadlines, and struggle to meet their savings targets due to the lack of a centralized and automated tracking system.

The proposed Budget Planning System provides a comprehensive digital solution that allows users to set monthly budgets for various expense categories such as food, transportation, education, healthcare, and entertainment. The system enables users to record daily transactions, view real-time summaries of their spending, compare actual expenditures against planned budgets, and receive alerts when they are approaching or exceeding their limits. The system also supports goal-based savings, allowing users to define financial targets and track their progress over time.

This project is developed using modern web technologies such as **HTML**, **CSS**, **JavaScript**, **PHP**, and **MySQL**, which provide a reliable, secure, and scalable platform for financial data management. The system can be accessed through desktop or mobile browsers connected to the internet. In the future, advanced features such as AI-based financial recommendations, integration with banking APIs, and automated report generation can be added to further enhance the system's capabilities. Overall, the Budget Planning System demonstrates the practical use of information technology in improving financial literacy and personal financial management.

OBJECTIVES OF THE PROJECT

The main objective of the **Budget Planning System** project is to design and develop a computerized system that helps users plan, track, and manage their personal or organizational finances efficiently. The system aims to provide a comprehensive, user-friendly, and secure platform where users can set budgets, record transactions, monitor expenses, and achieve their financial goals.

The specific objectives of the project are as follows:

1. To develop an online platform that allows users to create and manage monthly or annual budgets for various income and expense categories.
2. To provide a secure user registration and authentication system that protects personal financial data from unauthorized access.
3. To enable users to record daily transactions including income, expenses, and savings in a structured database.
4. To design a real-time dashboard that displays spending summaries, budget utilization, and financial health indicators through charts and graphs.
5. To implement an alert and notification system that warns users when their spending in a particular category approaches or exceeds the defined budget limit.
6. To support goal-based savings planning, allowing users to define financial targets and track their progress over a defined period.
7. To generate detailed financial reports such as monthly expense summaries, category-wise analysis, and savings trend reports for informed decision-making.
8. To minimize manual effort by automating calculations of totals, balances, and budget variances.
9. To ensure data security and confidentiality through proper authentication, authorization, and encrypted storage mechanisms.
10. To enhance financial literacy and awareness among users by providing meaningful insights into their income and spending patterns.

Overall, the objective of this project is to demonstrate the practical implementation of web development technologies and database management systems in solving real-world financial management problems for individuals and small organizations.

PROBLEM STATEMENT

In today's fast-paced world, managing personal and organizational finances effectively is one of the most important yet challenging tasks. Most individuals and families do not follow a structured approach to budget planning, which results in overspending, insufficient savings, and inability to meet financial goals. Traditional methods of budget tracking, such as maintaining handwritten notebooks, paper-based ledgers, or basic spreadsheet files, are highly inefficient and prone to human errors. These manual approaches fail to provide real-time insights into financial status and cannot alert users when their spending exceeds planned limits.

One of the major problems with existing manual budgeting methods is the lack of centralized data management. Financial records are scattered across multiple notebooks or files, making it difficult to get a comprehensive view of income, expenses, and savings at any given time. Users often lose track of their transactions, forget to record expenses, or make calculation mistakes that result in incorrect financial reports. This lack of accuracy and organization often leads to poor financial decision-making, debt accumulation, and failure to achieve savings targets.

Another significant issue is the absence of visual analytics and progress tracking in traditional budget management. Without graphical representations of spending patterns, category-wise expense breakdowns, and savings trends, users are unable to identify areas where they are overspending or underperforming financially. Moreover, there is no mechanism to set financial goals, monitor progress toward those goals, or receive timely reminders about upcoming bill payments or budget limits.

Therefore, there is a strong need for a computerized and automated **Budget Planning System** that can address these challenges by offering a secure, reliable, and user-friendly platform for comprehensive financial management. The proposed system aims to automate budget tracking, provide real-time financial insights, support goal-based savings, and generate detailed reports to help users make informed financial decisions and improve their overall financial well-being.

SCOPE OF THE PROJECT

The scope of the **Budget Planning System** project defines the boundaries, features, and functionalities of the system and explains how it can be applied in real-world financial management scenarios. The primary scope of this project is to develop a web-based application that helps individuals and small organizations create budgets, record transactions, monitor expenses, and achieve savings goals. The system focuses on improving the overall financial management process by replacing manual methods with an automated, accurate, and reliable digital solution.

The project covers essential functionalities such as user registration, login authentication, income and expense category management, monthly budget creation, daily transaction recording, budget utilization tracking, savings goal management, and automated financial report generation. The system also provides an administrative module that allows administrators to manage user accounts, view financial summaries, and oversee system operations. These features ensure smooth functioning of the platform and deliver meaningful value to all users.

An important aspect of the scope is the provision of data visualization and analytics tools. The system incorporates graphical dashboards displaying pie charts, bar graphs, and trend lines that help users understand their financial health at a glance. The report generation module supports monthly expense summaries, category-wise breakdowns, and savings progress reports that assist users in making data-driven financial decisions. This makes the system suitable not only for day-to-day expense tracking but also for long-term financial planning.

The scope of this project is primarily limited to the development of a web-based system using technologies such as HTML, CSS, JavaScript, PHP, and MySQL. The system is designed to operate on a local server or within a small business or educational environment. However, in the future, the system can be expanded to include advanced features such as integration with banking APIs, mobile application support, AI-powered financial advisory, multi-currency support, and cloud-based data storage. These enhancements can significantly increase the scalability, reach, and usability of the system.

METHODOLOGY / TECHNOLOGY USED

The development of the **Budget Planning System** follows a systematic and structured methodology to ensure the successful design, implementation, and testing of the application. The project is developed using the **Software Development Life Cycle (SDLC)** approach, which includes several stages such as planning, analysis, design, development, testing, implementation, and maintenance. This methodology helps in organizing project activities, minimizing development errors, and ensuring that the final system meets user requirements effectively.

In the **planning phase**, the project requirements were identified, and the objectives of the system were defined based on the limitations of traditional manual budget management methods. In the **analysis phase**, detailed information about user requirements, system functionalities, data processing needs, and security requirements was collected. During the **design phase**, system architecture, database structure, and user interface layouts were created using standard diagrams such as Data Flow Diagrams (DFD) and Entity Relationship (ER) diagrams. The **development phase** involved writing program code using appropriate programming languages and tools to implement all required system features.

The **testing phase** ensures that the system functions correctly and meets user expectations. Various types of testing, such as unit testing, integration testing, system testing, and user acceptance testing, were performed to identify and fix errors. After successful testing, the system is implemented and made available for use. The **maintenance** stage involves updating the system, fixing issues, and improving performance as needed over time.

Technologies Used in the Project

Front-End Technologies

HTML (HyperText Markup Language):

HTML is used to create the basic structure of web pages in the Budget Planning System. It defines elements such as forms, input fields, buttons, tables, and navigation menus that users interact with.

CSS (Cascading Style Sheets):

CSS is used to design and format the visual appearance of web pages. It enables the creation of attractive, responsive layouts with consistent color schemes, typography, and styling across all pages of the system.

JavaScript:

JavaScript is used to add interactivity, dynamic validation, and real-time calculations to the system. It powers features such as live budget progress bars, form validation, and interactive chart rendering.

Bootstrap:

Bootstrap is a front-end framework used to design responsive web pages that adapt seamlessly to different screen sizes including computers, tablets, and smartphones.

Chart.js:

Chart.js is a JavaScript library used to render interactive charts and graphs such as pie charts for expense categories, bar charts for monthly comparisons, and line charts for savings trends on the dashboard.

Back-End Technologies**PHP (Hypertext Preprocessor):**

PHP is a server-side scripting language used to handle user requests, process financial data, manage user sessions, implement business logic, and communicate with the database. It is the core technology driving the system's back-end operations.

MySQL:

MySQL is a relational database management system used to store and manage all application data including user profiles, budget categories, transaction records, savings goals, and financial reports. It ensures secure, organized, and efficient data storage and retrieval.

Development Tools

1. **Visual Studio Code:** Used as the primary code editor for writing, managing, and debugging program files.
2. **XAMPP Server:** Used to configure and run a local Apache web server and MySQL database environment during development and testing.
3. **Web Browser (Google Chrome):** Used to test, run, and debug the web application during development.
4. **phpMyAdmin:** A web-based tool used for designing database schemas, managing tables, and executing SQL queries.
5. **Operating System:** Windows 10 or higher, used for system development and execution.

Overall, the methodology and technologies used in this project ensure the development of a reliable, efficient, and user-friendly Budget Planning System that meets modern financial management needs and demonstrates the practical application of web development concepts learned during the BCA program.

HARDWARE AND SOFTWARE REQUIREMENTS

The successful development and implementation of the **Budget Planning System** requires appropriate hardware and software components to ensure smooth operation, efficient performance, and reliable system functionality. The hardware provides the physical infrastructure needed to run the application, while the software provides the tools and environment required for development, execution, and maintenance.

Hardware Requirements

The hardware requirements for developing and running the Budget Planning System are as follows:

1. **Processor:** Intel Core i3 or higher — A capable processor is required to execute program instructions efficiently and support multitasking during system development and runtime operations.
2. **Random Access Memory (RAM):** Minimum 4 GB (Recommended: 8 GB) — Adequate RAM ensures smooth performance of the system while simultaneously running development tools, web servers, and database management systems.
3. **Hard Disk / Storage:** Minimum 500 GB — Sufficient storage is required for software installations, database files, project documents, backups, and transaction data.
4. **Monitor:** LCD / LED Monitor — A display device is required to view the graphical user interface, program code, dashboard charts, and system output.
5. **Keyboard and Mouse:** Standard input devices required for data entry, navigation, and development tasks.
6. **Network Connection:** Internet / Local Area Network (LAN) — A stable network connection is required for testing, accessing online resources, and enabling multi-user access.
7. **Printer (Optional):** May be used to print financial reports, budget summaries, and project documentation when needed.

Software Requirements

The software requirements for developing and operating the Budget Planning System include the following:

1. **Operating System:** Windows 10 or higher
Provides the platform on which the application and all development tools run.
2. **Programming Languages:** HTML for web page structure;
 - **CSS** for styling and responsive design;
 - **JavaScript** for client-side interactivity and chart rendering;
 - **PHP** for server-side processing and business logic implementation.
3. **Database Management System (DBMS):** MySQL
Used to store and manage all financial data including users, budgets, categories, transactions, savings goals, and reports in a structured and relational format.
4. **Web Server:** Apache Server (XAMPP)
Responsible for hosting the application locally and processing all HTTP requests from users.
5. **Web Browser:** Google Chrome / Mozilla Firefox / Microsoft Edge
Required to access, test, and interact with the Budget Planning System interface.
6. **Development Tools:** Visual Studio Code for code editing;
 - **XAMPP** for local server configuration;
 - **phpMyAdmin** for database management and schema design.
7. **Chart.js Library:** Used for rendering interactive financial charts and graphs on the user dashboard.

Overall, the above hardware and software requirements ensure that the Budget Planning System operates efficiently, supports accurate financial data processing, and provides a reliable environment for system development and deployment.

EXPECTED OUTCOME

The **Budget Planning System** is expected to provide an efficient, reliable, and user-friendly platform for managing personal and organizational finances. The successful implementation of this system will simplify the process of budget creation and expense tracking for users, while eliminating the limitations and inaccuracies of traditional manual financial management methods. It will help ensure better financial discipline, reduce unnecessary spending, and support informed financial decision-making.

One of the most significant expected outcomes of this project is the effective centralization and management of all financial data through a structured relational database system. The system will securely store user profiles, budget categories, transaction records, savings goals, and historical financial data, allowing easy access, retrieval, and analysis. This will enable users to maintain a complete and accurate financial history and make data-driven decisions about their spending and savings.

Another important expected outcome is improved financial awareness and control through an interactive dashboard with charts, graphs, and real-time summaries. Users will be able to visualize their income versus expense ratios, category-wise spending distributions, and savings progress at a glance. The automated alert system will notify users when their expenditure in any category is approaching or exceeding the defined budget limit, helping them stay within their financial plans.

The Budget Planning System is also expected to deliver meaningful analytical reports, including monthly financial summaries, category-wise expense breakdowns, and savings trend analysis. These reports will help users evaluate their financial performance, identify areas for improvement, and plan future budgets more accurately. In addition, the system is designed to be scalable for future enhancements such as mobile application support, multi-currency management, banking API integration, and AI-powered financial advisory features. Overall, this project will demonstrate the practical application of web development and database technologies in solving real-world financial management challenges.

CONCLUSION

The **Budget Planning System** project has been successfully designed and developed to provide an efficient, automated, and comprehensive digital solution for managing personal and organizational finances. The system replaces traditional manual budgeting methods with a modern web-based platform that ensures accuracy, transparency, and real-time visibility into financial activities. It provides a user-friendly interface that allows individuals and small organizations to plan budgets, record transactions, track expenses, and monitor savings goals with ease and efficiency.

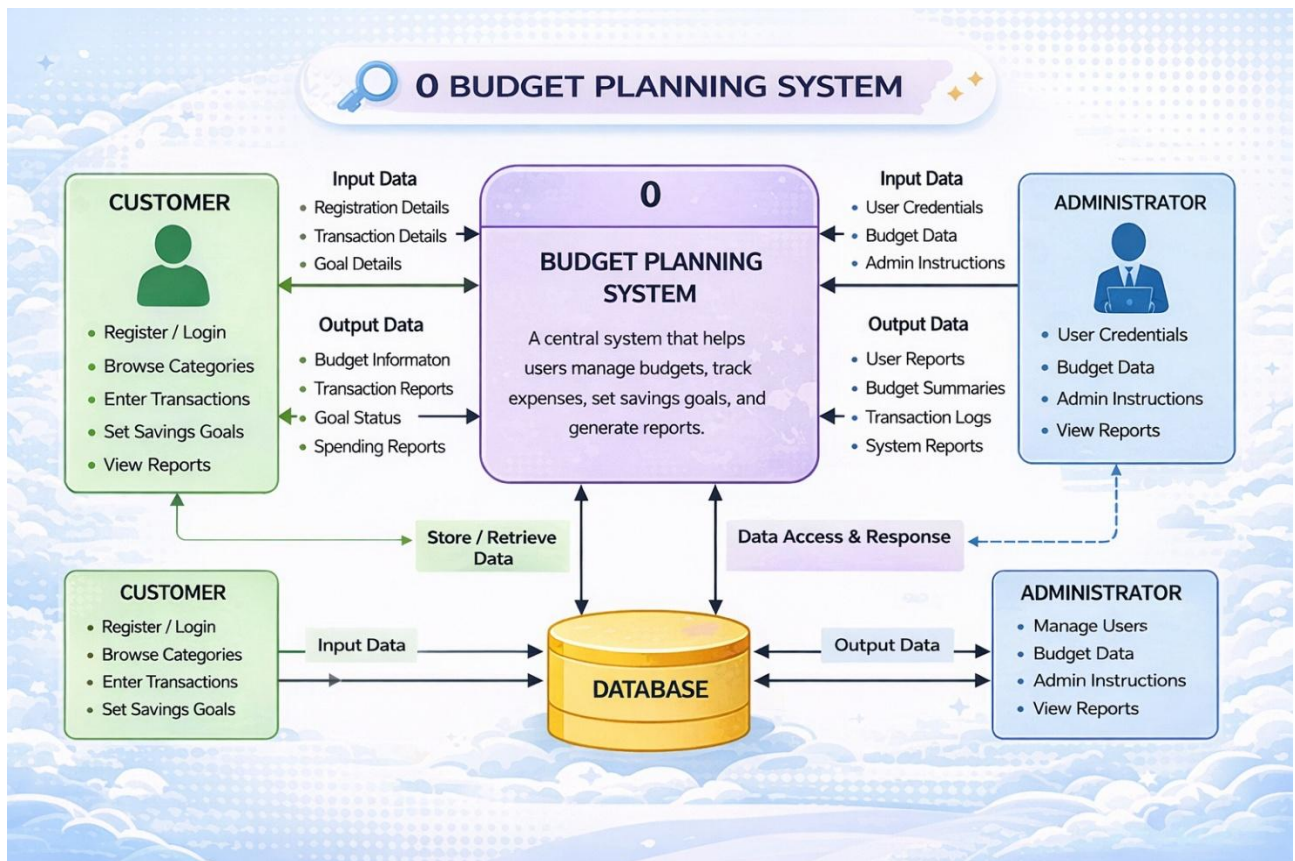
This project has demonstrated the practical application of web development technologies and database management systems acquired during the Bachelor of Computer Applications (BCA) program. The system successfully implements all planned features including user authentication, budget category management, daily transaction recording, real-time dashboard analytics, automated alerts, goal-based savings tracking, and comprehensive financial report generation. The integration of Chart.js for visual analytics enhances the user experience by presenting financial data in an intuitive and informative graphical format.

The implementation of the Budget Planning System contributes directly to improved financial literacy and discipline among users. By providing clear visual feedback on spending patterns, budget variances, and savings progress, the system empowers users to make well-informed financial decisions and work consistently toward their financial goals. The secure database management ensures the confidentiality and integrity of sensitive financial data, which is essential for maintaining user trust in a financial application.

In the future, the system can be significantly enhanced by incorporating features such as mobile application development, integration with banking APIs for automatic transaction import, AI-powered spending recommendations, multi-currency support, and cloud-based data storage. These improvements will greatly expand the system's capabilities and applicability for larger user bases. Overall, the project fulfills all its defined objectives and highlights the important role of information technology in promoting financial well-being and efficient financial management in the modern world.

DATA FLOW DIAGRAM (DFD) – LEVEL 0

The **Data Flow Diagram (DFD) Level 0**, also known as the **Context Diagram**, represents the overall structure of the **Budget Planning System** as a single process. It shows how the system interacts with external entities — primarily the User (individual or organizational budget planner) and the Administrator. This diagram provides a high-level view of the system and illustrates the flow of data between users and the system without showing internal processes in detail.

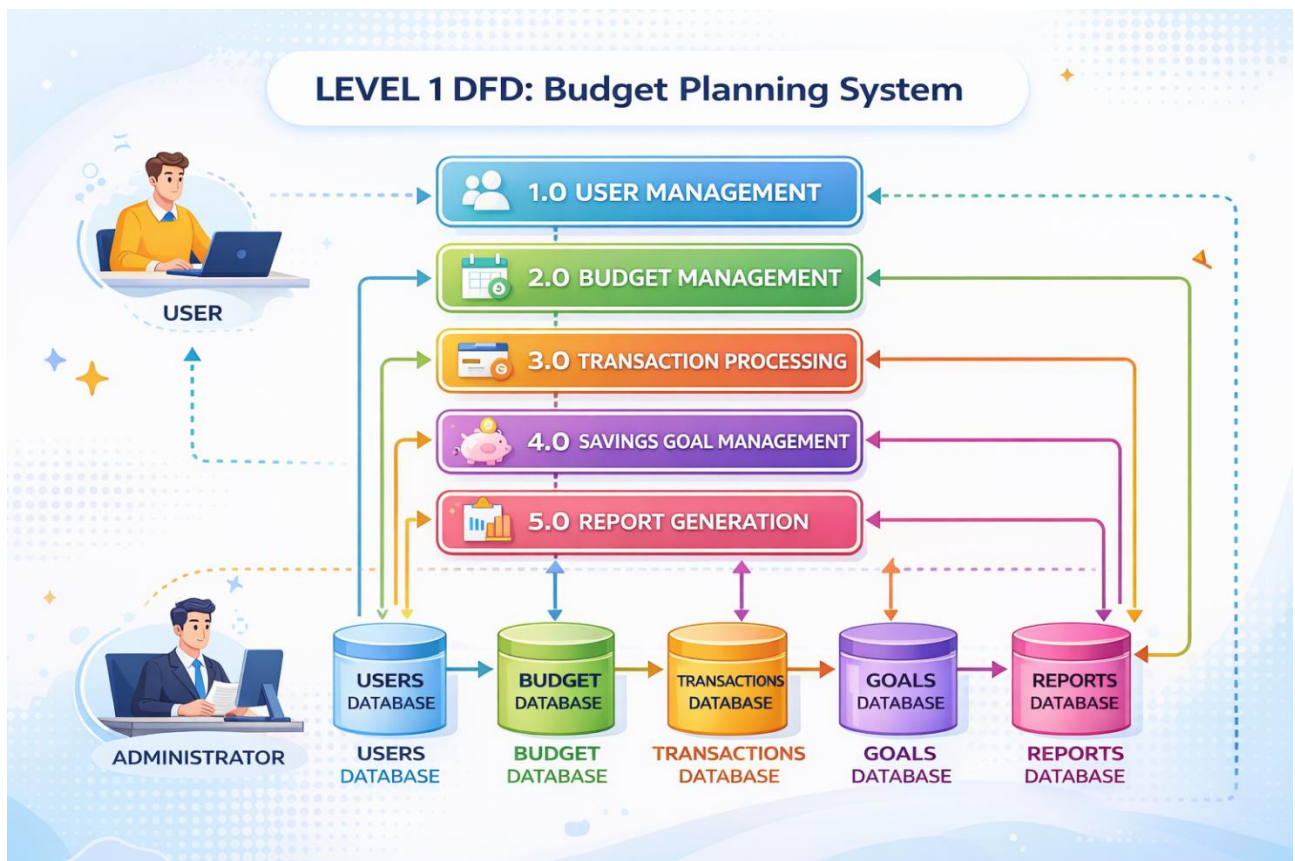


This process represents the entire Budget Planning System that handles all financial management activities including user registration, budget creation, transaction recording, report generation, and savings goal monitoring.

DATA FLOW DIAGRAM (DFD) – LEVEL 1

The **Data Flow Diagram (DFD) Level 1** provides a detailed breakdown of the main process of the **Budget Planning System** into smaller sub-processes. It explains how data moves between different components of the system, including users, administrators, and databases. This level reveals the internal working of the system and shows how each major function is performed step by step.

The Level 1 DFD divides the main system into several key processes: User Management, Budget Management, Transaction Processing, Savings Goal Management, Report Generation, and Alert Management. Each process interacts with specific data stores to store and retrieve information as required. The diagram illustrates the logical flow of financial data through the system and helps developers design each component with clarity.

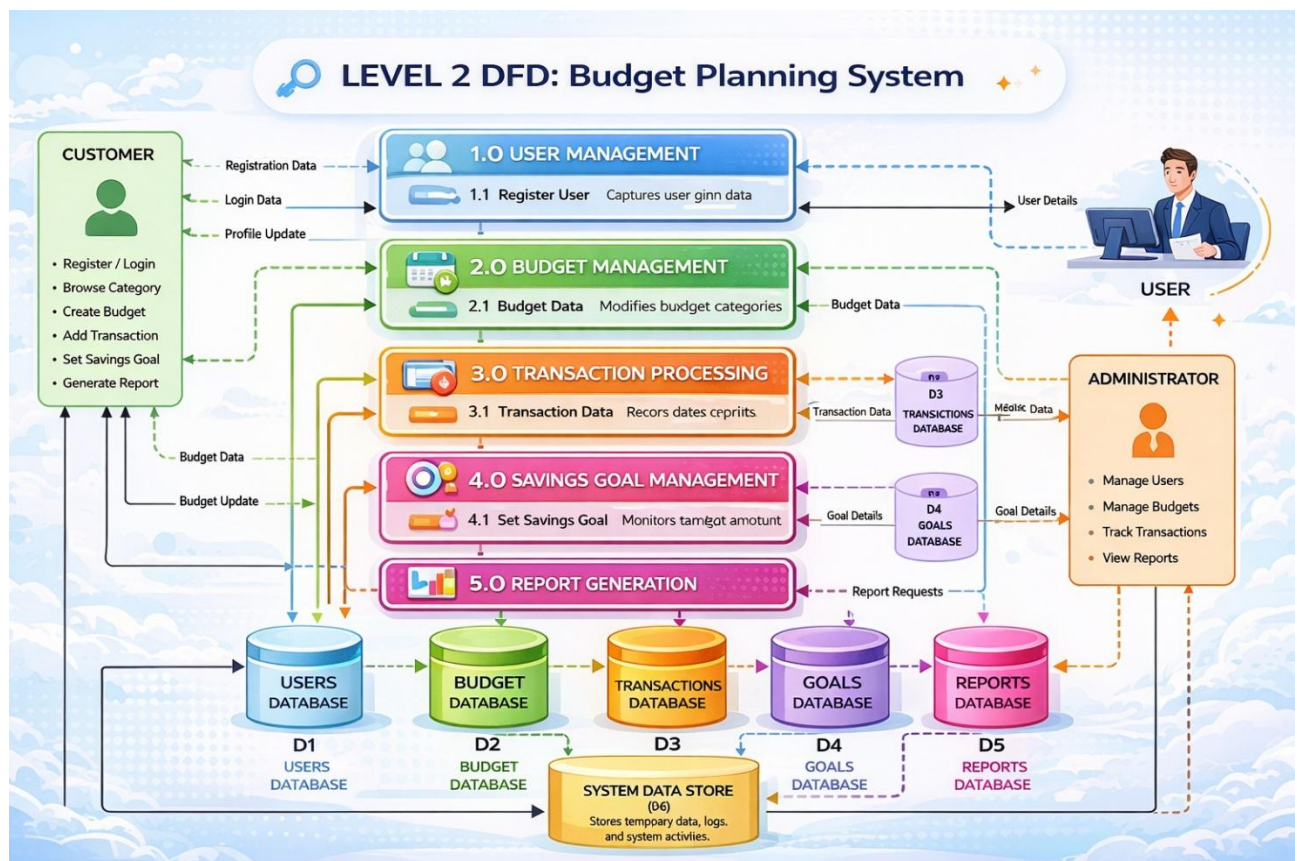


The Level 1 Data Flow Diagram provides a clear and detailed representation of how the Budget Planning System operates internally. It shows the interaction between processes, data stores, and external entities, and helps ensure proper system design and correct implementation of all functional modules.

DATA FLOW DIAGRAM (DFD) – LEVEL 2

The **Data Flow Diagram (DFD) Level 2** provides a more detailed view of the internal processes of the **Budget Planning System**. It breaks down each major process from Level 1 into smaller sub-processes to clearly show how data flows within the system. This level helps developers and stakeholders understand the step-by-step operations performed by the system in detail.

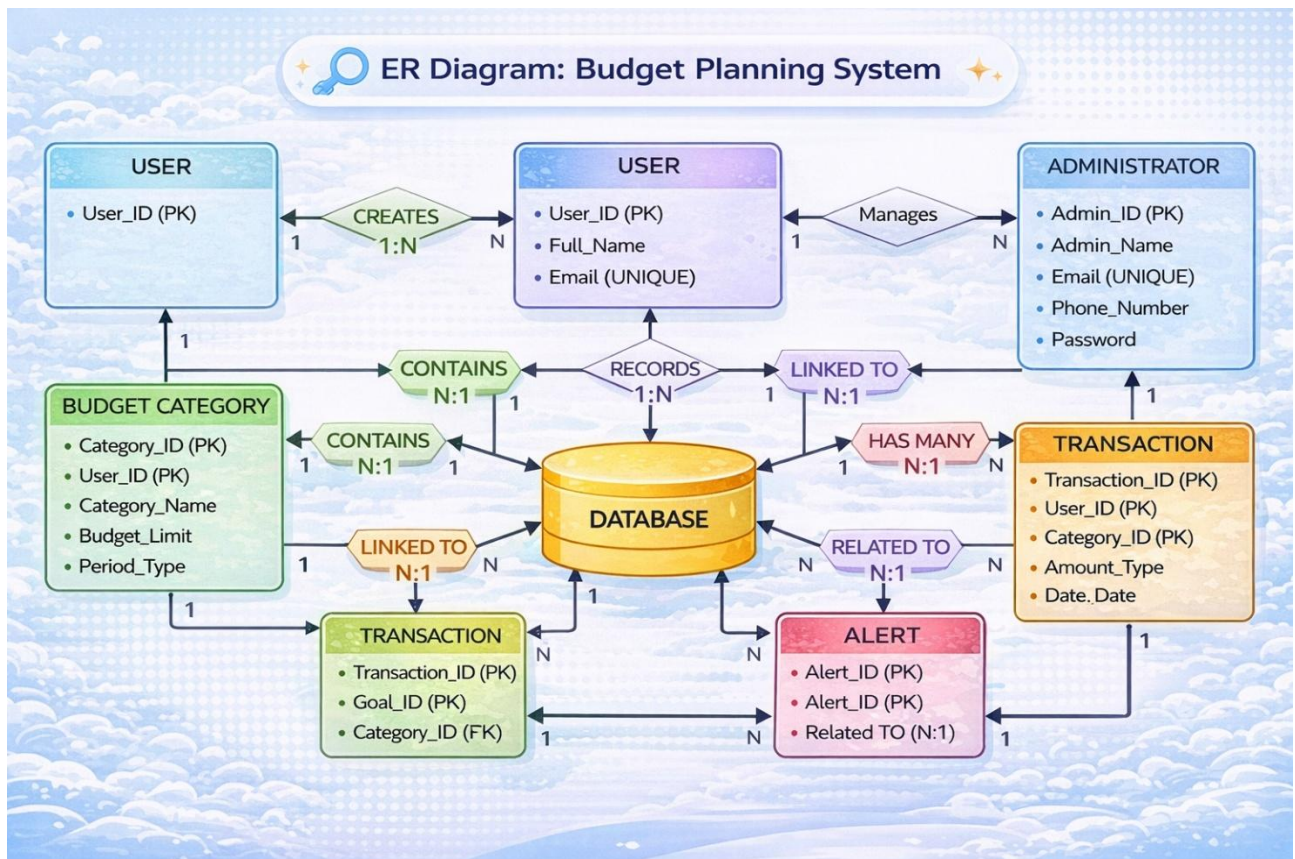
The Level 2 DFD explains how users interact with the system to register, log in, create budgets, record transactions, set savings goals, and receive reports and alerts. It also shows how the administrator manages user accounts, budget categories, and system reports. Each sub-process is connected to specific data stores where financial information is stored and retrieved.



The Level 2 Data Flow Diagram provides a comprehensive and detailed representation of the internal operations of the Budget Planning System. It explains how each functional module is implemented and how data flows between sub-processes and databases, ensuring proper system design, efficient data management, and smooth overall functionality.

ER DIAGRAM (ENTITY RELATIONSHIP DIAGRAM)

In the Budget Planning System, the main entities include **User**, **BudgetCategory**, **Transaction**, **SavingsGoal**, **Alert**, and **Administrator**. These entities interact with each other to perform various financial operations such as recording transactions, managing budget categories, tracking savings, and generating alerts. The ER Diagram ensures data consistency, minimizes redundancy, and supports efficient data retrieval across the system.



The Entity Relationship Diagram provides a clear understanding of how different entities in the Budget Planning System are interconnected. It helps developers design a well-structured, normalized database and ensures smooth and reliable system operation. The ER Diagram plays a vital role in maintaining data accuracy, integrity, and consistency throughout all financial management operations of the system.

ENTITIES, ATTRIBUTES, AND RELATIONSHIPS

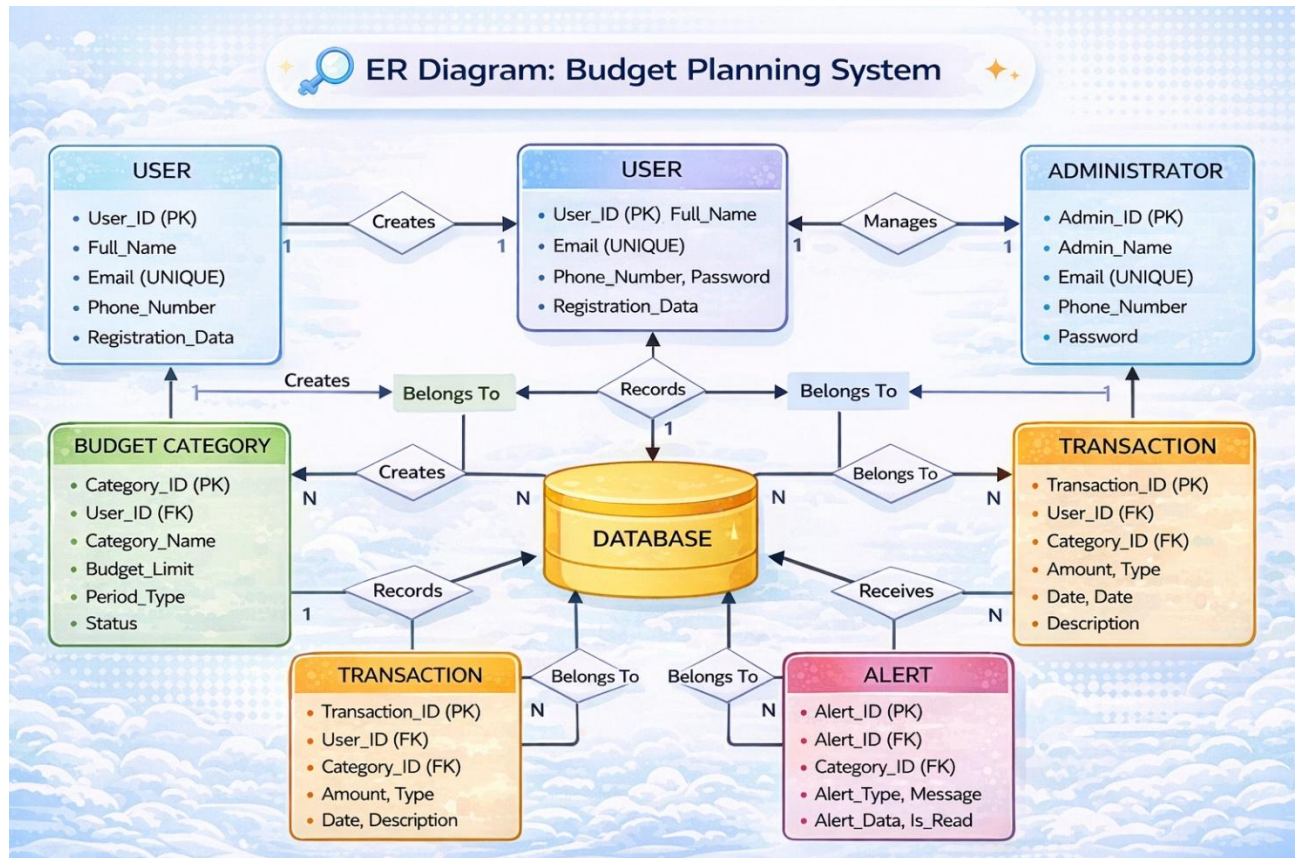
1. Entities

An Entity represents a real-world object or concept about which data is stored in the system. In the Budget Planning System, the main entities are:

1. User
2. BudgetCategory
3. Transaction
4. SavingsGoal
5. Alert
6. Administrator

2. Attributes of Entities

Attributes are the properties or characteristics of an entity. Key attributes for each entity are listed below:



3. Relationship Summary

- One USER can create many BUDGET CATEGORIES. (1:N)
- One USER can record many TRANSACTIONS. (1:N)
- One BUDGET CATEGORY can contain many TRANSACTIONS. (1:N)
- One USER can set many SAVINGS GOALS. (1:N)
- One USER can receive many ALERTS. (1:N)
- One ADMINISTRATOR can manage many USERS. (1:N)

DATABASE STRUCTURE

The database structure defines the tables used in the system along with their fields, data types, sizes, and key constraints. It ensures proper data organization, efficient storage, and accurate retrieval for reliable system performance. The Budget Planning System uses six main tables as described below.

1. USER TABLE

Field Name	Data Type	Size	Key	Description
User_ID	INT	10	PK	Unique user identifier
Full_Name	VARCHAR	100		User's full name
Email	VARCHAR	100	UNIQUE	Login email address
Phone_Number	VARCHAR	15		Contact number
Password	VARCHAR	255		Encrypted login password
Registration_Date	DATE	—		Date of account creation

2. BUDGET CATEGORY TABLE

Field Name	Data Type	Size	Key	Description
Category_ID	INT	10	PK	Unique category ID
User_ID	INT	10	FK	Reference to User
Category_Name	VARCHAR	100		E.g., Food, Transport
Budget_Limit	DECIMAL	10,2		Monthly budget amount
Period_Type	VARCHAR	20		Monthly / Yearly
Status	VARCHAR	20		Active / Inactive

3. TRANSACTION TABLE

Field Name	Data Type	Size	Key	Description
Transaction_ID	INT	10	PK	Unique transaction ID
User_ID	INT	10	FK	Reference to User
Category_ID	INT	10	FK	Reference to Budget Category
Amount	DECIMAL	10,2		Transaction amount
Type	VARCHAR	10		Income / Expense
Date	DATE	—		Date of transaction
Description	TEXT	—		Optional transaction note

4. SAVINGS GOAL TABLE

Field Name	Data Type	Size	Key	Description
Goal_ID	INT	10	PK	Unique savings goal ID
User_ID	INT	10	FK	Reference to User
Goal_Name	VARCHAR	100		Name of the savings goal
Target_Amount	DECIMAL	10,2		Total amount to be saved
Saved_Amount	DECIMAL	10,2		Amount saved so far
Target_Date	DATE	—		Goal completion deadline
Status	VARCHAR	20		In Progress / Achieved

5. ALERT TABLE

Field Name	Data Type	Size	Key	Description
Alert_ID	INT	10	PK	Unique alert ID
User_ID	INT	10	FK	Reference to User
Category_ID	INT	10	FK	Reference to Budget Category
Alert_Type	VARCHAR	50		Warning / Exceeded
Message	TEXT	—		Alert message content
Alert_Date	DATETIME	—		Date and time of alert
Is_Read	TINYINT	1		0 = Unread, 1 = Read

6. ADMIN TABLE

Field Name	Data Type	Size	Key	Description
Admin_ID	INT	10	PK	Unique admin ID
Admin_Name	VARCHAR	100		Administrator name
Email	VARCHAR	100	UNIQUE	Admin email address
Phone_Number	VARCHAR	15		Contact number
Password	VARCHAR	255		Encrypted login password

There are a total of 6 tables in the Budget Planning System: User, BudgetCategory, Transaction, SavingsGoal, Alert, and Admin. Primary Keys ensure unique identification of each record, while Foreign Keys maintain relational integrity between tables. This structured and normalized database design supports efficient financial data processing, secure user data management, and accurate generation of financial reports.

REFERENCES / BIBLIOGRAPHY

The successful completion of the Budget Planning System project was made possible with the help of various books, websites, and online resources. These references provided valuable knowledge about programming, database management, web development, financial systems, and software engineering concepts used in this project.

Books

1. **Korth, H. F., & Silberschatz, A.**
Database System Concepts
McGraw-Hill Education
2. **Elmasri, R., & Navathe, S. B.**
Fundamentals of Database Systems
Pearson Education
3. **Pressman, Roger S.**
Software Engineering: A Practitioner's Approach
McGraw-Hill Education
4. **Sommerville, Ian**
Software Engineering
Pearson Education
5. **Welling, L., & Thomson, L.**
PHP and MySQL Web Development
Addison-Wesley

Websites

1. www.w3schools.com
2. www.javatpoint.com
3. www.geeksforgeeks.org
4. www.tutorialspoint.com
5. www.mysql.com
6. www.chartjs.org

These references were used to understand system development concepts, database design principles, programming techniques, financial management theory, and implementation methods required to successfully design and develop the Budget Planning System project.

INTERNSHIP DETAILS (MANDATORY)

Company Name	ARDENT COMTUTECH PVT. LTD
Company Address	MODULE 103, SDF BUILDING, SECTOR V, SALT LAKE, KOLKATA -700091
Email	internship@ardentsoftware.in
Contact Number	033-40073507
Duration of Internship	60 Days / 8 Weeks
Internship Start Date	14 SEPTEMBER 2025
Internship End Date	23 NOVEMBER 2025
Supervisor Name	SOURAV GOSWAMI
Contact	8945531907
Nature of Work / Training Details	ONLINE

INTERNSHIP CERTIFICATE

Industrial Internship Certificate



**ARDENT
COMPUTECH
PVT. LTD.**



This certificate is awarded to

AKASH KUMAR

of

NETAJI SUBHAS UNIVERSSSITY, JAMSHEDPUR

for successfully completing the **Industrial Internship** on

DATA SCIENCE, ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING USING PYTHON

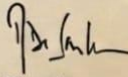
from **ARDENT COMPUTECH PVT. LTD.**

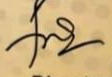
NOVEMBER 13, 2025 to DECEMBER 23, 2025

and implementing the project titled

GLOBAL MOBILE PHONE PRICE ANALYSIS – 2025

Certificate ID: **ARDENT/178190**
Issue Date: **DECEMBER 23, 2025**


**Director
Technology Services**


**Director
Operations**








*Thank
you!*